Switchblade: tools and frameworks explained

The store itself

Tool	What it does for Switchblade	Think of it as
Next.js	Builds the website customers see and the logic behind it: pages, bag, checkout, account	The shop building and its staff
React	Makes the pages interactive: size selection, bag updates, filters	The shop's fittings and displays
TypeScript	Catches coding mistakes before they reach customers	A checklist the builders follow
Payload CMS	The admin panel where the team adds products, prices, banners, size charts and coupons, without a developer	The back office
Payload hooks	When the team edits something, the site updates within seconds	Staff changing the window display instantly
Kill switches	Lets the team pause COD or checkout with one click during an emergency, no developer needed	The shop's shutter

Speed and protection

Tool	What it does for Switchblade	Think of it as
Vercel	Hosts the website on servers in Mumbai and serves pages from locations close to each customer	The shop's premises, in a prime location
Vercel Firewall	Blocks attacks and bad bots before they reach the store	Security at the entrance
Cloudflare DNS	Points switchblade.in to the right place	The shop's address

Tool	What it does for Switchblade	Think of it as
Cloudflare Images	Stores product photos and sends each phone the right size and format, so pages load fast on 4G	A photo studio that prints every photo in the perfect size
Cloudflare Turnstile	Checks that people logging in are humans, not bots, with no annoying puzzles	A guard who spots fake customers
Upstash Redis	Stops anyone asking for too many OTPs, blocks double-clicks from creating two orders, and remembers delivery answers for an hour so checkout stays fast	A doorman with a notepad: "You've tried 5 times, please wait," and "Yes, we deliver to 500081"

Data and accounts (Supabase)

Tool	What it does for Switchblade	Think of it as
Supabase Postgres	The main database: products, stock, orders, customers, coupons	The master ledger
Supabase Auth	Phone OTP login and keeping customers logged in securely	The membership desk
Supabase Storage	Keeps invoices and return photos	The filing cabinet
Supavisor pooler	Manages database connections so sale-day traffic doesn't overwhelm it	A queue manager at the ledger desk
Supabase Queues	Holds tasks like "send WhatsApp" or "book courier" safely until they're done, retrying if they fail	A to-do list that never loses a task
pg_cron	Runs scheduled jobs: releases unpaid holds, checks payments and shipments, makes backups	An alarm clock for routine chores
Edge Functions	The workers that carry out queued tasks: sending messages, booking couriers	Back-office staff
Daily backups	A copy of all data every day, kept 7 days	A photocopy of the ledger each night

Tool	What it does for Switchblade	Think of it as
Staging project	A full copy of the store for testing changes before customers see them	A rehearsal stage
Point-in-time recovery (growth)	Restores the database to any exact second	Rewinding the ledger
Read replica (scale)	A second copy of the database that handles browsing traffic	A second ledger desk on busy days

Payments, delivery and messages

Tool	What it does for Switchblade	Think of it as
Razorpay Magic Checkout	One-tap checkout with saved addresses, coupons, UPI, cards, COD, and COD risk checks to cut fake orders	The billing counter
Shiprocket	Checks if a pincode is deliverable, books couriers, tracks parcels, handles return pickups	The delivery partner
WhatsApp (via Gupshup or Interakt)	Sends order confirmed, shipped and delivered updates	Calling the customer with updates
Resend	Sends receipts and invoices by email from switchblade.in	The post office

Watching over the store

Tool	What it does for Switchblade	Think of it as
Sentry	Alerts the team the moment anything breaks, showing exactly where	A fire alarm that tells you which room
Uptime robot	Checks every minute that the site is up, calls the team if it's down, and shows customers a status page	A night watchman plus a notice board
Vercel Speed Insights	Measures how fast pages load for real customers	A stopwatch on every visit
PostHog	Shows where customers drop off between product and payment, with session replays	CCTV for the shopping journey

Tool	What it does for Switchblade	Think of it as
Log drain (growth)	Keeps searchable history of everything that happened	The shop's diary

Building and releasing safely

Tool	What it does for Switchblade	Think of it as
GitHub	Stores all the code, with the full history of every change	The blueprint archive
GitHub Actions	Automatically checks every change before release	An inspector before opening
Vercel preview deploys	Every change gets its own test link to review before going live	A trial display before the real window
Playwright checkout test	A robot shopper runs through checkout on every change and alerts if it breaks	A mystery shopper
k6 load test (growth)	Simulates thousands of shoppers before a big sale	A fire drill for sale day
Inngest or Trigger.dev (growth)	Manages complex multi-step workflows as the business grows	A project manager for long tasks